



香 港 大 學

THE UNIVERSITY OF HONG KONG

COMP7705

**MSc(CompSc) Interim Report and
Presentation**

**An innovative AI-powered interview assessment tool
for company**

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1. Executive Summary

This Interim Report presents the progress of our team in developing an AI-powered virtual interview assessment tool. We have successfully completed in core text-based system and made significant advancement in **website performance**.

The system now features a complete **role-based platform** with three dedicated portals: **Candidate**, **Company (HR)**, and **Admin**. It supports end-to-end AI interview flow, voice interaction, and comprehensive reporting. A testing version is live at <https://218.102.191.34:8080/FYP/labour/>.

2. Project Objectives

The primary objectives are to:

- Develop a basic AI tool that automates resume screening by matching candidate qualifications to job descriptions, reducing human effort
- Generate customized interview questions based on job types
- Analyze candidate responses for relevance and quality, providing a ranking score to shortlist top applicants
- Demonstrate how AI can minimize bias and improve efficiency in hiring, while learning practical skills in Python, web apps, and data processing.

3.Milestones

Tasks	Estimated completion time	Estimated number of learning hours
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1. Research & Planning — Review AI Hiring PDF and related literature — Define objectives, scope, and stream (Multimedia Computing) — Install Python, Streamlit, OpenAI/Groq API — Draft and submit Detailed Project Proposal — Create and initialize project webpage on school account	March 10, 2026	60 hours
2. Basic Interface & Question Generation — Build simple chat interface using Streamlit — Implement basic question generation with LLM API (prompt engineering) — Test with sample job descriptions — Update project webpage with screenshots and progress	April 7, 2026	50 hours
3. Multi-turn Conversation & Initial Scoring — Add conversation history management for multi-turn interviews — Develop initial automated scoring (relevance, technical depth) — Conduct preliminary tests with dummy responses — Update webpage with demo video/screenshots	May 5, 2026 (Progress Update 2)	45 hours

4. Advanced Features & Interim Report — Implement full scoring dimensions (clarity, value alignment) — Add intelligent follow-up probes — Generate basic reports and recommendations - Prepare and deliver Interim Report & Presentation	June 1, 2026 (Interim Report & Presentation)	50 hours
5. Testing, Refinement & Bias Check - Full system end-to-end testing with 20–30 dummy cases - Check for bias and improve prompts - Fix bugs and optimize performance Update webpage with test results	June 16, 2026 (Progress Update 3)	70 hours
6. Final Polish & Oral Preparation - Refine UI and report format - Conduct final full tests and collect results - Prepare demo video and oral examination materials - Final webpage update	July 6, 2026 (Progress Update 4)	50 hours
7. Final Submission - Complete Project Report - Submit final webpage (July 13)	July 17, 2026 (Project Report)	30 hours

4. Current Implementation Progress

4.1 Role-Based Portals and Flows

To provide a clear and user-friendly experience for different stakeholders, the platform has been designed with three distinct role-based portals: Candidate Portal, Company / Interviewer Portal, and Admin Portal. Each portal has its own dedicated workflow, interface, and functionality, ensuring that every user type can perform their tasks efficiently:

Candidate Portal (This portal is designed for job seekers):

1. Landing page → Login / Registration
2. Dashboard – view applied jobs and match status
3. CV / Profile Upload – drag-and-drop CV submission with AI parsing
4. Job Browsing & Matching – browse available positions and see AI match scores
5. System Check Page – camera preview, microphone test, and network check
6. AI Interview Room – enter the live interview with voice interaction and virtual avatar
7. Interview Completion – receive confirmation message (scores are hidden for fairness)

Company / Interviewer Portal (This portal is for HR professionals and hiring managers):

1. Dashboard – overview of open job posts, question bank setting and candidate statistics
2. Job Creation Page – create new job posts, define requirements and core values
3. Question Bank Management – manage fixed questions and trigger AI-generated questions
4. Candidates List & Matching Page – view all applicants with AI match scores and filters
5. AI Interview Report Review – access full conversation history, multidimensional scores, strengths/weaknesses, and hiring recommendation
6. Hiring Decision – shortlist, reject, or schedule next round for candidates

Admin Portal (This portal is for IT/platform administrators):

1. Admin Dashboard – system overview and quick statistics
2. AI Provider & Model Settings – configure Gemini, Grok/xAI, OpenAI or local models, test connections
3. User & Role Management – manage accounts, roles, and permissions
4. Monitoring & Logs – view API latency, error rates, usage analytics, and security logs
5. Governance & Compliance – policy management and fairness review

4.2 Four AI Agent Architecture

To achieve a modular, maintainable and scalable AI interview system, we have designed and implemented a multi-agent architecture. Each agent is a specialized module with clearly defined roles, inputs, outputs, and prompt templates. This design allows different parts of the interview process to be developed, tested, and improved independently.

The four core agents are:

- **Profile & Job Understanding Agent:** This agent is responsible for extracting structured information from unstructured candidate CVs/profiles and job posts. It converts raw text into clean JSON data, including candidate skills, experience, education, and job requirements. This structured data serves as the foundation for subsequent matching and interview stages. This agent is preferred used in LLM trained with RAG due to the personal data privacy.
- **Matching Agent:** This agent compares the candidate's structured profile against the job requirements and company core values. It calculates an overall match score, identifies strengths and gaps, and provides a recommendation (Strongly Recommend / Recommend / On Hold / Not Recommended) along with clear explanations for the interviewer.
- **Interview Agent:** This is the core conversational agent. It conducts the live interview by selecting appropriate questions from the question bank, asking intelligent follow-up questions based on the candidate's previous answers, and maintaining a natural, professional flow. It decides in real time whether to ask a follow-up, move to the next question, or end the interview. This agent is preferred to using the gemini AI model which have the low latency and real time voice to handle the interview conversation.

- **Evaluation Agent:** After the interview is completed, this agent analyzes the full transcript. It scores the candidate across multiple dimensions (job relevance, technical depth, communication clarity, problem-solving approach, and company value alignment), provides evidence-based strengths and weaknesses, generates a detailed report, and gives a final hiring recommendation with fairness warnings if any potential bias is detected.

All agents communicate using strict JSON input/output formats, making the system reliable and easy to extend. Prompt templates for each agent are stored in the AGENTS/ folder for easy maintenance and future improvement.

5. Next Steps and Future Work

The team has successfully completed and established a solid foundation with a fully functional text-based AI interview system, intelligent follow-up logic, role-based portals (Candidate, Company, and Admin), and automatic evaluation reports shown in figure (Part 7).

We are now entering the multimodal phase. The following items are prioritized in preparation final submission:

- **Virtual Interviewer Avatar** Implement a realistic avatar with lip-sync capability and integrate it smoothly into the interview room interface.
- **Camera System + Anti-Deepfake Protection** Add camera-enabled interview room with liveness detection, deepfake prevention, and continuous identity verification.
- **Facial Expression Analysis** Integrate MediaPipe to detect and analyze smile intensity, tension, and eye contact as supplementary behavioral signals.
- **Voice Tone Analysis** Incorporate prosodic features (speech rate, pause patterns, intonation) to complement content-based scoring.

- Comprehensive Scoring Report Combine text answers, facial expressions, voice tone, and authenticity signals into one final multimodal evaluation report with visual charts and hiring recommendations.

6.Project Website

The project webpage is live at <https://wp2025.cs.hku.hk/msp25020/> and has been regularly updated with screenshots, progress updates, and live demo links.

7.Figures

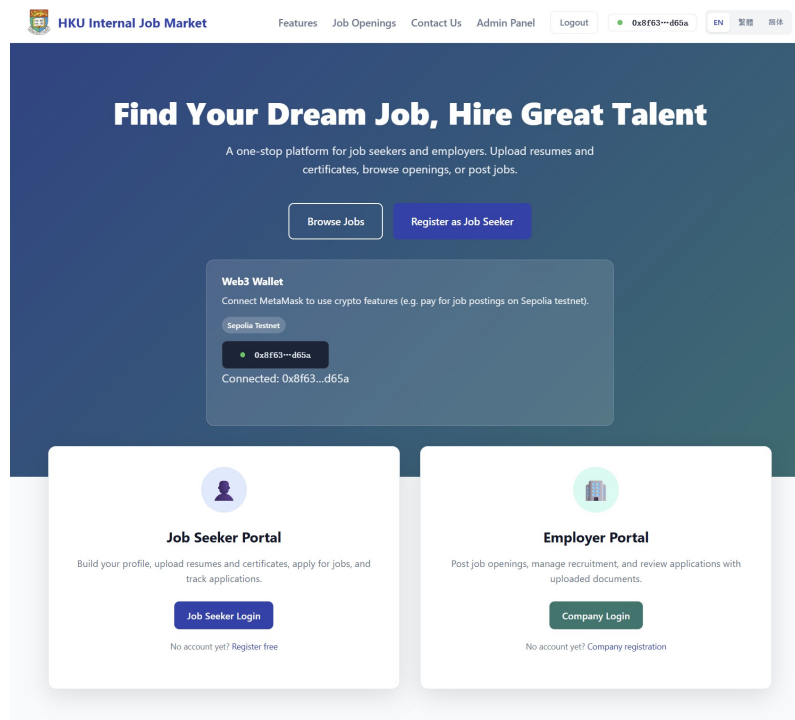


Figure 1. Home page interface with two separate portal login options (e.g., Job Seeker Portal and Employer Portal).

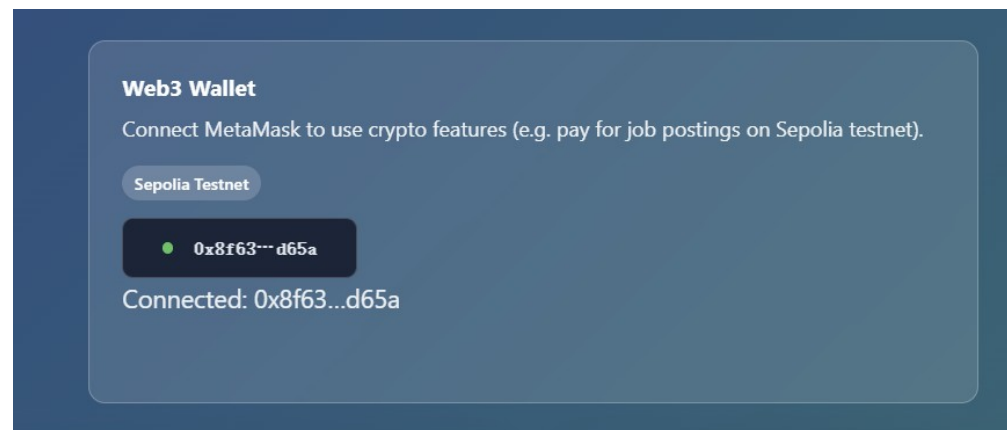


Figure 2. Web3 wallet with commercial features, such as requiring connected wallet and cryptocurrency payment per job recruitment post.

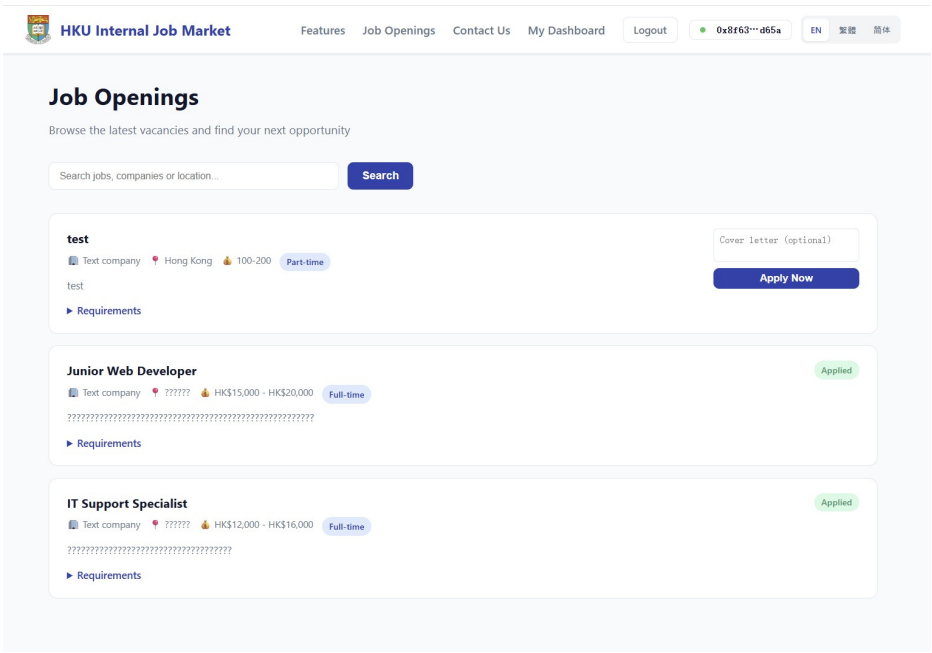


Figure 3. Job Openings section, allowing users to browse current vacancies and discover their next career opportunity.

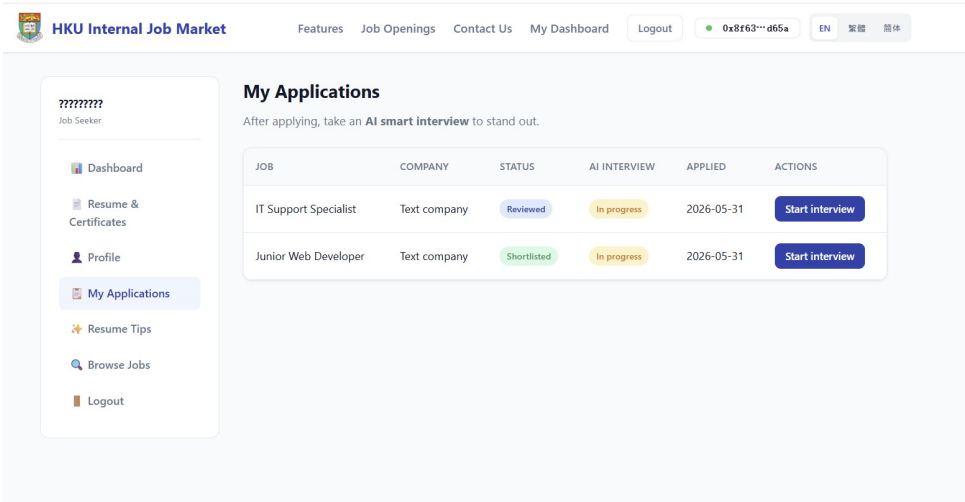


Figure 4. Job seeker dashboard — view submitted applications, start an AI-powered interview, and track job status.

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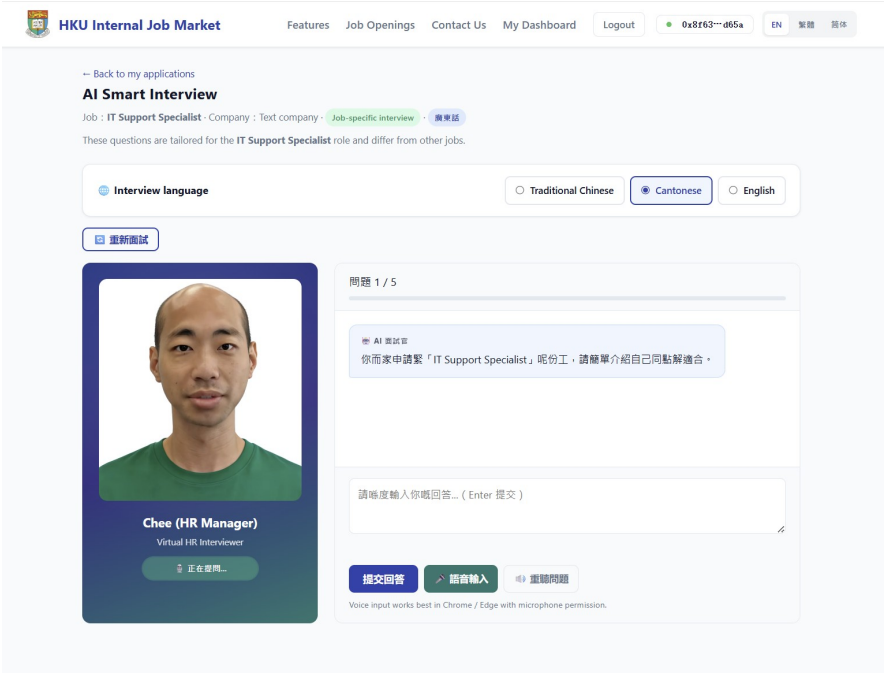


Figure 5. Initial AI interview interface — note that this component is subject to significant future updates.

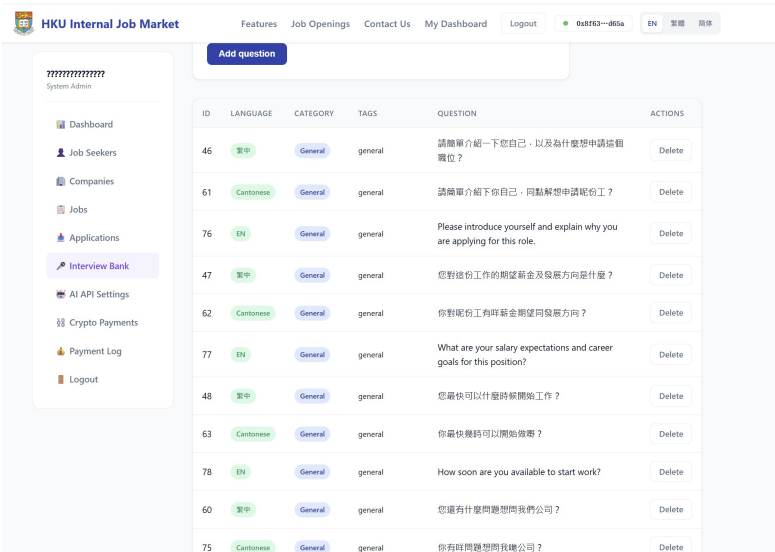


Figure 6. AI-generated questions available in English, Mandarin, and Cantonese.

